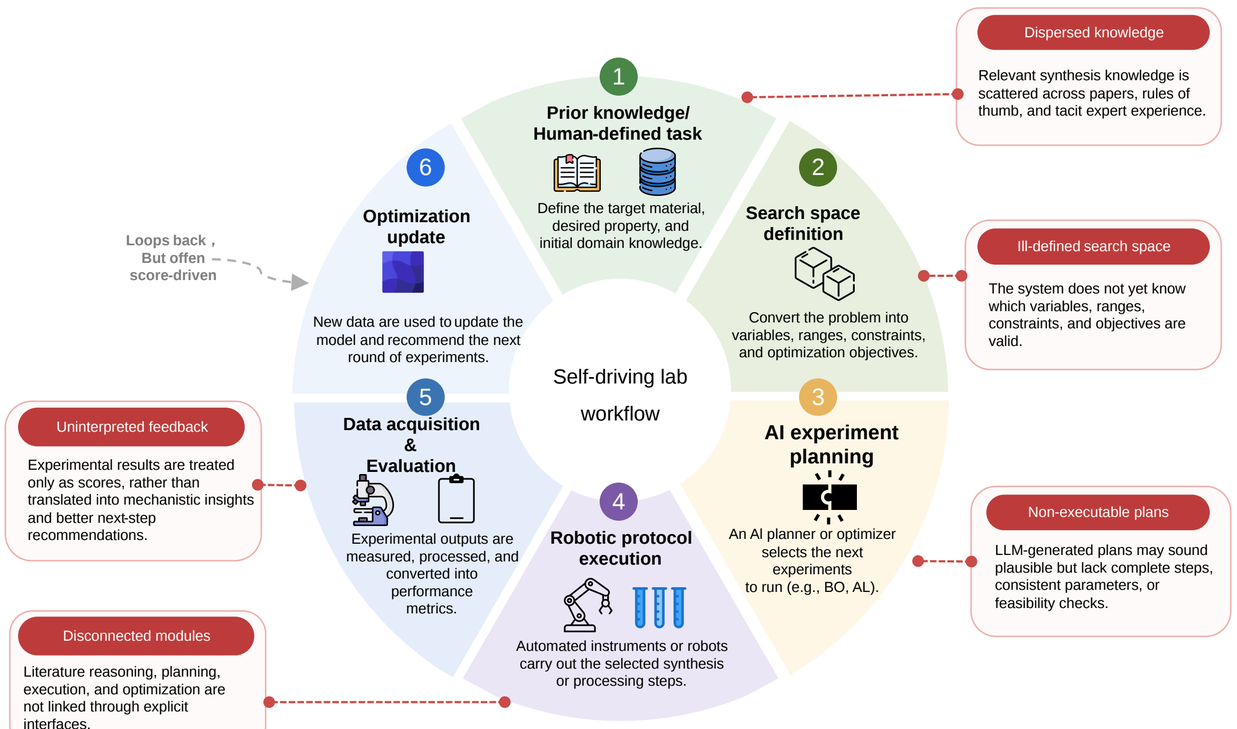
a

Traditional self-driving lab workflow



Dispersed knowledge

Relevant synthesis knowledge is scattered across papers, rules of thumb, and tacit expert experience.

Ill-defined search space

The system does not yet know which variables, ranges, constraints, and objectives are valid.

Non-executable plans

LLM-generated plans may sound plausible but lack complete steps, consistent parameters, or feasibility checks.

Uninterpreted feedback

Experimental results are treated only as scores, rather than translated into mechanistic insights and better next-step recommendations.

Disconnected modules

Literature reasoning, planning, execution, and optimization are not linked through explicit interfaces.

Bottlenecks across the entire pipeline lead to low efficiency, high cost, and slow iteration.

b

SupraMAS closes the decision loop

1 Strategy Tree Construction

Organize literature-derived strategies, constraints and expected outcomes into a recursive strategy tree to build executable design spaces.

Dispersed Knowledge (Literature, Rules, Constraints) → **Strategy Tree** (Executable Design Space)

2 Template-constraint Expert-Critic Loop

Iteratively refine candidate ideas and experimental protocols. Expert ensures scientific consistency; Critic ensures procedural feasibility and executability.

Expert (Scientific Consistency) ⇄ Iterative refinement ⇄ **Critic** (Feasibility & Execution) → **Executable Protocol**

Closed Decision Loop

3 Mechanism-informed Bayesian Optimization

Combine mechanistic priors with experimental feedback to efficiently explore the parameter space.

Mechainstic Priors (Theory & Empirical) + **Experimental Feedback** → **BO with Weighted Priors** (Mechanism-guided efficient search)

4 Hierarchical Backtracking

Trace experimental outcomes back through experiment,idea and strategy levels. Resolve conflicting mechanisms within a constrained hypothesis space and recommend actionable adjustments.

Hierarchical backtracking — Backtracking & Update — **Hypothesis Space** (Resolve Conflicts)

SupraMAS closes the decision loop by integrating structured knowledge, expert critique, optimization, and hierarchical backtracking.

c

SupraMAS implementation workflow

